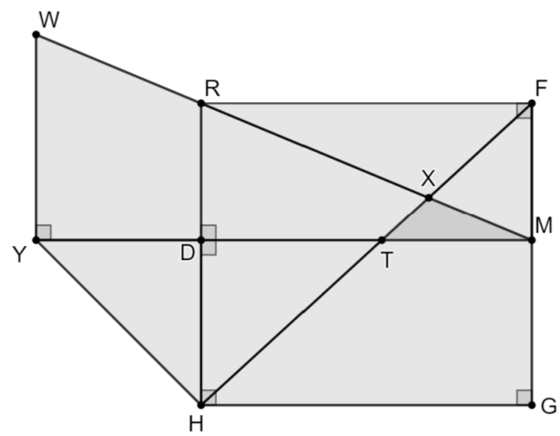
Geometry	Name Answer Key
Unit 7	Date________________
Lesson 9 Practice	Period_________

Alafia River State Park has a set of mountain bike trails. Mario draws out an area of the trails that he plans to ride. His diagram is shown.

The table shows known measurements of the trails.

$m\angle DYH = 45°$	$YW = 5$ kilometers
$m\angle XMT = 23°$	$DM = 8$ kilometers
$HD = 4$ kilometers	$\triangle FMR \cong \triangle DRM$
$m\angle GHF = 43°$	



Complete the table. In the work column, write the equation or trigonometric ratio used to determine the side measure. Round to the nearest tenth.

	Side	Work	Side Measure
1.	$\overline{HF}$	$cos\,43 = \frac{8}{HF}$	10.9
2.	$\overline{HY}$	$sin\,45 = \frac{4}{HY}$	5.7
3.	$\overline{RD}$	$tan\,23 = \frac{RD}{8}$	3.4
4.	$\overline{RM}$	$cos\,23 = \frac{8}{RM}$	8.7

5. Mario starts his ride at point W and rides along the perimeter of the park. What is the total distance, in kilometers, Mario rides? Round to the nearest tenth of a kilometer.

$$tan\,43 = \frac{GF}{8}; GF = 8\,tan\,43\,; GF = 7.46$$

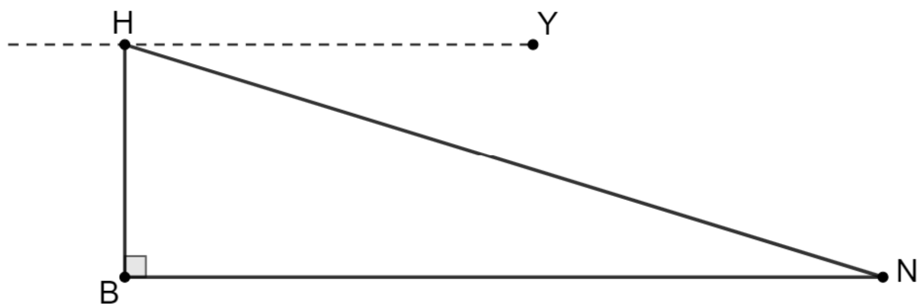
$$YD = (5.7)^2 - 4^2; YD = 4; YM = 4 + 8 = 12$$

$$WM = \sqrt{5^2 + 12^2} = 13; RW = WM - RM; RW = 13 - 8.7 = 4.3$$

$$5 + 5.657 + 8 + 7.406 + 8 + 4.309 = 38.4 \text{ km}$$

6. What is the total distance, in kilometers, if Mario rides from W to Y to H to T to X to F to R ending at W. Round to the nearest tenth of a kilometer.
$5 + 5.657 + 10.939 + 8 + 4.309 = 33.9$ km

Gabby is flying a drone over Lake Jackson. The drone is represented by point H and is 265 feet above Lake Jackson which is represented by point B. The angle of depression, $\angle YHN$, is the angle that Gabby will fly the drone for it to land on the fishing pier, represented by point N. The measure of $\angle YHN$ is $17°$.



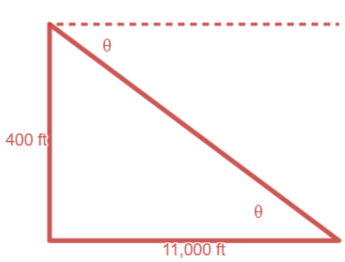
7. Explain why $\angle YHN$ is congruent to $\angle BNH$.
$\overline{HY}$ is parallel to the ground, $\overline{BN}$. $\overline{HN}$ is a transversal to parallel line segments $\overline{HY}$ and $\overline{BN}$, making alternate interior angles $\angle YHN$ and $\angle BNH$ congruent.

8. Determine the horizontal distance, BN, the drone is from the fishing pier. Round your answer to the nearest tenth.
$tan\,17 = \frac{265}{BN}$; $BN = \frac{265}{BN} \approx 866.8$ ft

9. Determine the distance, HN, the drone must travel to reach the fishing pier. Round your answer to the nearest tenth.
$sin\,17 = \frac{265}{HN}$; $HN = \frac{265}{sin\,17} \approx 906.4$ ft

10. Zac is also flying a drone over Lake Jackson. His drone is $11{,}000$ feet across Lake Jackson horizontally from the fishing pier. If the height of his drone is 400 feet, what is the angle of depression Zac should fly his drone so it lands on the fishing pier? Round your answer to the nearest tenth of a degree.

$$tan\,\theta = \frac{400}{11000}$$



$$\theta = tan^{-1}\frac{400}{11000} \approx 2.082 \approx 2.1°$$